

MATH 314 – EXAM 1 STUDY GUIDE

Lecture 1 – Intro to Stats (Homework 1 in General)

- Populations and Samples (p. 6-10)
- Categorical vs. Numerical data (p. 12-18)
- Parameters vs. Statistics (p. 20-33)
 - Proportion
 - Mean
 - Median
 - Percentiles/Quartiles
- Measures of Spread (p. 34-40)
 - Variance & standard deviation
 - Interquartile range
- Plots in RStudio (p. 42-60)
 - Tables
 - Bar charts
 - Histograms
 - Box plots
 - Scatter plots
 - Density plots
- Comparative Studies (p. 62-72)
 - Explanatory & Response Variables
 - Observational & Experimental Studies
 - Confounding Variables

Lecture 2 – Intro to Probability

- Sample Space & Events (p. 4-6)
- Set Operations (p. 8-12)
 - Union
 - Intersection
 - Complement
 - Difference
- Probability of Events (p. 15-21)
 - Using R operators (&, |, !)
- Sets and Probability (p. 23-25)

- Discrete Uniform Example
- Conditional Probability (p. 27-40)
 - $P(B|A) = P(A \cap B) / P(A)$
 - Solving for conditional probabilities given certain information
 - Law of Total Probability
 - Bayes' Theorem (briefly)
 - **Homework 2, Q2**
- Independence (p. 42-44)
 - $P(A \cap B) = P(A)P(B)$ (If A and B are pairwise independent)
 - **Homework 2, Q3-Q5**
- Association Rules (p. 46-51)
 - For the rule $\{B\} \rightarrow \{A\}$ (B implies A):
 - Support: $P(A \cap B)$
 - Confidence: $P(A|B)$
 - Lift: $P(A|B)/P(A)$
 - **Homework 2, Q6**
 - NOTE: Also be aware of how to use association rules given a table like the one seen in this section of the notes.

Lecture 3 – Random Variables

- Discrete Random Variables (p. 8-10)
- Probability Mass Function (PMF) (p.11-12, 15-16)
- Cumulative Distribution Function (CDF) – Discrete (p.13-14)
 - Be able to solve statements like $P(X \leq 3)$ given a table of possible values and probabilities.
 - Note that $P(X < 3)$ and $P(X \leq 3)$ are NOT the same for discrete cases
- Probability Density Function (PDF) (p.17-18, 20-21)
- Cumulative Distribution Function (CDF) – Continuous (p.19)
 - Note that $P(X < 3)$ and $P(X \leq 3)$ ARE the same for continuous cases
 - The probability of $P(X = x)$ is 0 for any x, for continuous cases.
- Expected Value (p.27-30)
 - **Homework 3, Q1 & Q4**
- Variance/Standard Deviation (p.31-32)
- Distributions (p.34-42)
 - Bernoulli

- Uniform (Discrete & Continuous)
- Exponential
 - Use of pexp as the CDF
 - **Homework 3, Q3**
- Normal (Lec03_Examples)
 - Use of pnorm as the CDF
 - **Homework 3, Q2**
 - 68-95-99.7 Rule
 - z-score
- Note: Be aware of how these are structured—you may see other distributions beyond these that follow similar properties
- Parameters of distributions (p.44-50)

Lecture 4 - Likelihood

- Likelihood function structure (p. 10, 12)
- Maximum likelihood (p. 14-15)
- Logarithm rules (p. 16)
- Log likelihood function (p. 17-18)
 - **Homework 4, Q1**
- Derivative rules (p. 19-20)
- Computer Examples with optim (p. 24-29)
 - **Homework 4, Q2**

Log-Likelihood Tips

- Look for terms that can cancel, combine, or be taken outside product/summation notation.
- Remember common log and derivative properties.
- You will probably end up with the sum of x_i in some form.
- Beware the order of functions, parentheses, and exponents when translating to RStudio.
- Remember that distributions with **two parameters** must have an argument that contains a **vector of both values** when writing their log-likelihood functions in RStudio.